# The University of Texas at Austin
# Homework Assignment 11

*Predictive Analytics*

November 13, 2026

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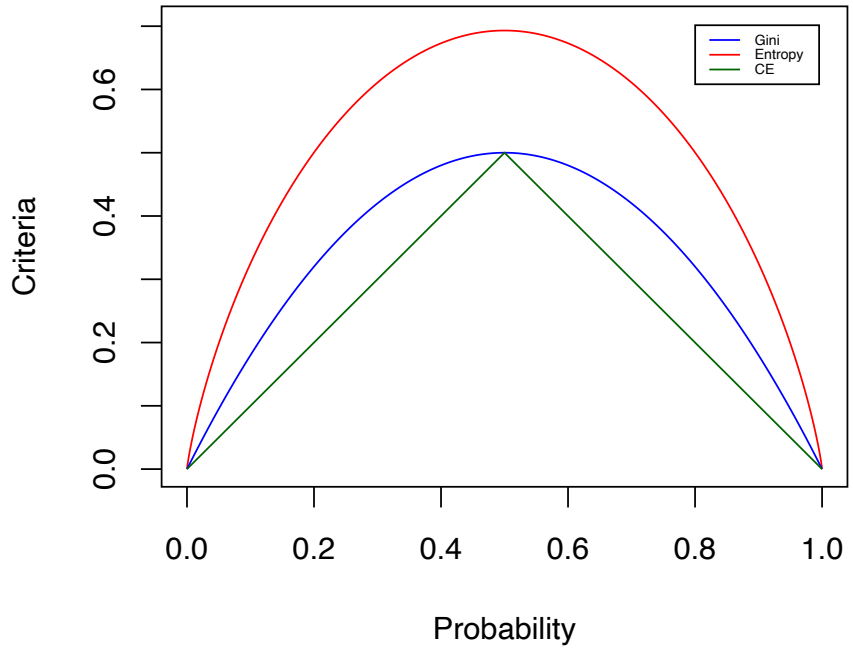
**Instructions:** Provide your complete solution to the following problems. Final answers only, without appropriate justification, will receive zero points even if correct.

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## Classification Trees.

**Problem 11.1.** (10 points) Solve Problem **8.4.3** (pp.361-362) in the textbook.

**Solution.** Here is the plot:




**Problem 11.2.** (10 points) A classification tree is constructed to predict whether a student will pass *Predictive Analytics*. There are two categorical predictors:

- $X_1$ indicating whether the student passed *Linear Algebra* prior to enrolling in *Predictive Analytics*, and
- $X_2$ indicating whether the student passed *Mathematical Statistics* prior to enrolling in *Predictive Analytics*.

Here is the table from a data set of students:

| $X_1$ | $X_2$ | outcome |
|---|---|---|
| 0 | 0 | 5 passed, 20 didn't |
| 1 | 0 | 10 passed, 10 didn't |
| 0 | 1 | 15 passed, 20 didn't |
| 1 | 1 | 15 passed, 5 didn't |

What's the first split made using the Gini index? Be careful to calculate the **weighted** average.

**Solution.** Since both $X_1$ and $X_2$ are binary, we can make the first split in a unique way with respect to $X_1$ or with respect to $X_2$. If we split along $X_1$, we have that

- 20 passed and 40 didn't;
- 25 passed and 15 didn't.

So, the Gini index is

$$\frac{60}{100} \cdot 2 \cdot \left(\frac{20}{60}\left(1 - \frac{20}{60}\right)\right) + \frac{40}{100} \cdot 2 \cdot \left(\frac{25}{40}\left(1 - \frac{25}{40}\right)\right) = 0.4541667$$

If we split along $X_2$, we have that

- 15 passed and 30 didn't;
- 30 passed and 25 didn't.

So, the Gini index is

$$\frac{45}{100} \cdot 2 \cdot \left(\frac{15}{45}\left(1 - \frac{15}{45}\right)\right) + \frac{55}{100} \cdot 2 \cdot \left(\frac{30}{55}\left(1 - \frac{30}{55}\right)\right) = 0.4727273$$

Since the first Gini index is smaller, we should split along $X_1$.

**Problem 11.3.** (30 points) A classification tree is fitted to be used to classify drivers into one of three categories: *Good, Medium* and *Bad*. There are three terminal nodes in the classification tree designating the regions $R_1, R_2$ and $R_3$. Here is the breakdown of categories in each region:

| Region | Good | Medium | Bad |
|---|---|---|---|
| $R_1$ | 70 | 20 | 10 |
| $R_2$ | 30 | 50 | 10 |
| $R_3$ | 20 | 25 | 35 |

Calculate the overall classification error rate, Gini index, and cross-entropy using **weighted** averages.

**Solution.** There are $n_1 = 100$ observations in $R_1$, $n_2 = 90$ observations in $R_2$, and $n_3 = 80$ observations in $R_3$. The total number of observations is $n = 270$. So, in each calculation, the weight for $R_1$ is $w_1 = \frac{100}{270} = \frac{10}{27}$, the weight for $R_2$ is $w_2 = \frac{90}{270} = \frac{1}{3}$, and the weight for $R_3$ is $w_3 = \frac{80}{270} = \frac{8}{27}$.

- In $R_1$, the most common category is *Good*, so the classification error rate equals

$$\frac{20 + 10}{100} = 0.3.$$

In $R_2$, the most common category is *Medium*, so the classification error rate equals

$$\frac{30+10}{90} = \frac{4}{9}.$$

In $R_3$, the most common category is *Bad*, so the classification error rate equals

$$\frac{20+25}{80} = 0.5625.$$

Overall, we obtain

$$\frac{10}{27} \cdot \frac{3}{10} + \frac{1}{3} \cdot \frac{4}{9} + \frac{8}{27} \cdot \frac{9}{16} = 0.4259259.$$

Equivalently, and way more easily, we could have calculated

$$\frac{20+10+30+10+20+25}{100+90+80} = \frac{115}{270} = 0.4259259.$$

- In $R_1$, the Gini index is

$$\frac{70}{100} \cdot \frac{30}{100} + \frac{20}{100} \cdot \frac{80}{100} + \frac{10}{100} \cdot \frac{90}{100} = 0.46$$

In $R_2$, the Gini index is

$$\frac{30}{90} \cdot \frac{60}{90} + \frac{50}{90} \cdot \frac{40}{90} + \frac{10}{90} \cdot \frac{80}{90} = 0.5679012$$

In $R_3$, the Gini index is

$$\frac{20}{80} \cdot \frac{60}{80} + \frac{25}{80} \cdot \frac{55}{80} + \frac{35}{80} \cdot \frac{45}{80} = 0.6484375$$

Overall, we have

$$\frac{10}{27}(0.46) + \frac{1}{3}(0.5679012) + \frac{8}{27}(0.6484375) = 0.5518004$$

- In $R_1$, the cross-entropy is

$$-(0.7\ln(0.7) + 0.2\ln(0.2) + 0.1\ln(0.1)) = 0.8018186.$$

In $R_2$, the cross-entropy is

$$-\left(\frac{30}{90}\ln\left(\frac{30}{90}\right) + \frac{50}{90}\ln\left(\frac{50}{90}\right) + \frac{10}{90}\ln\left(\frac{10}{90}\right)\right) = 0.9368883.$$

In $R_3$, the cross-entropy is

$$-\left(\frac{20}{80}\ln\left(\frac{20}{80}\right) + \frac{25}{80}\ln\left(\frac{25}{80}\right) + \frac{35}{80}\ln\left(\frac{35}{80}\right)\right) = 1.07173$$

Overall, we have

$$\frac{10}{27}(0.8018186) + \frac{1}{3}(0.9368883) + \frac{8}{27}(1.07173) = 0.9268156.$$